CHAPTER 10

WAVE OPTICS

CHAPTER ESSENCE AT A GLANCE:
Huygens' wave front construction, coherent sources, Young's double slit interference fringes, and single slit Fraunhofer diffraction.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Huygens' Wavefront Principle:** Every point on a primary wavefront acts as a secondary spherical wavelet source; the forward envelope tangent to all wavelets forms the subsequent wavefront.

• **Coherence & Two-Slit Interference:** Two coherent light sources with constant phase differences produce alternating bright and dark fringes on a distant screen through wave superposition.

• **Equidistant Fringe Width:** Double slit interference fringes maintain perfectly uniform spacing across the central pattern, directly proportional to screen distance and wavelength.

• **Diffraction vs Interference:** Interference arises from superposition of waves from two distinct coherent slits, whereas diffraction results from mutual interference between wavelets across parts of the same wavefront.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\beta = \frac{\lambda D}{d}$$

Fringe Width in Young's Double Slit Interference

$$w_0 = \frac{2\lambda D}{a}$$

Linear Width of Central Maximum in Single Slit Diffraction

$$a\sin\theta = n\lambda \quad (n = 1, 2, 3 \dots)$$

Condition for Secondary Minima in Single Slit Diffraction

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

Refraction at Plane Interface via Wave Theory

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Interference fringes are of EQUAL width and intensity; Diffraction central maximum is TWICE as wide as secondary fringes and intensity falls off rapidly.

• **Alert:** [Valuation Rubric] Draw two coherent slits S_1, S_2 with separation d, screen at distance D, path difference .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Wavefront: definition and types (spherical, cylindrical, plane wavefronts) [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Wavefront Definition]** A wavefront is the continuous locus of all points in a medium that oscillate in the same phase of disturbance at any given instant. A ray of light is always perpendicular to the wavefront. [1 Mark]

• **[Three Types of Wavefronts]** 1. Spherical Wavefront: Produced by a point source of light in an isotropic medium (concentric expanding spheres). 2. Cylindrical Wavefront: Produced by a linear source like a thin glowing slit (concentric expanding cylinders). 3. Plane Wavefront: A small portion of a spherical or cylindrical wavefront originating from a source at infinite distance (e.g. parallel wavefronts from the Sun). [1 Mark]

• **Key Exam Result:** Wavefront: locus of constant phase; Spherical (point), Cylindrical (slit), Plane (infinite)

2. Huygens' Principle of wave propagation [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Huygens' Principle Postulates]** 1. Every point on a given primary wavefront acts as a fresh source of secondary spherical disturbances called secondary wavelets, traveling in all directions with the speed of light in that medium. [1 Mark]

• 2. The forward envelope (common tangential surface) enclosing these secondary wavelets at any later time t gives the new position of the wavefront at that instant. (Wavelets do not propagate backward due to obliquity factor (1 + cos θ) divided by 2). [1 Mark]

• **Key Exam Result:** Every wavefront point acts as secondary wavelet source; forward tangent gives new wavefront

3. Proof of laws of reflection using Huygens' wave theory [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Diagram]** Plane wavefront AB incident at angle i on a reflecting plane surface XY. Let v be wave speed. Time taken by point B to reach surface at C is t = BC divided by v. In the same time t, secondary wavelet from point A expands to sphere of radius AD = v t. [1 Mark]

• **[Geometric Congruence Proof]** Draw tangent CD from C to wavelet sphere. CD represents reflected plane wavefront at angle r. Consider right-angled triangles ΔABC and ΔADC: 1. angle ABC = angle ADC = 90° (rays perpendicular to wavefronts). 2. Side BC = Side AD = v t. 3. Common hypotenuse AC = AC. Therefore, ΔABC approx. ΔADC by RHS congruence. Corresponding angles must be equal: angle BAC = angle DCA implies i = r. This rigorously proves the law of reflection! [2 Marks]

• **Key Exam Result:** Law of Reflection proved: Angle of incidence i = Angle of reflection r

4. Proof of laws of refraction using Huygens' wave theory (Snell's law) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Diagram]** Plane wavefront AB incident at angle i on refracting interface XY separating medium 1 (speed v_1) from medium 2 (speed v_2). Time taken by disturbance to travel from B to C in medium 1: t = BC divided by v_1. In same time t, secondary wavelet from A travels into medium 2 a distance AD = v_2 t. [1 Mark]

• **[Proof of Snell's Law]** Tangent CD represents refracted plane wavefront at angle r. From right triangle ΔABC: sin i = BC divided by AC = (v_1 t) divided by AC. From right triangle ΔADC: sin r = AD divided by AC = (v_2 t) divided by AC. Dividing both equations: sin i divided by sin r = (v_1 t divided by AC) divided by (v_2 t divided by AC) = v_1 divided by v_2. By definition of refractive index: n_{21} = v_1 divided by v_2 = n_2 divided by n_1. Therefore: sin i divided by sin r = n_2 divided by n_1 implies n_1 sin i = n_2 sin r. Proves Snell's law! [2 Marks]

• **Key Exam Result:** Snell's Law proved: sin i divided by sin r = v_1 divided by v_2 = n_2 divided by n_1

5. Principle of superposition of waves and resultant intensity formula I = I 1 + I 2 + 2√(I 1 I 2)cos\ϕ [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Superposition & Intensity Derivation]** Two waves $y_1 = a_1 \cos(\omega t)$ and $y_2 = a_2 \cos(\omega t + \Phi)$ meeting at a point. Resultant wave: $y = y_1 + y_2 = A \cos(\omega t + \theta)$. Resultant amplitude: $A^2 = a_1^2 + a_2^2 + 2 a_1 a_2 \cos \Phi$. [1.5 Marks]

• **[Resultant Intensity Formula]** Since intensity I proportional to A^2: $I = I_1 + I_2 + 2 \text{ sqrt}(I_1 I_2) \cos \Phi$. If two sources are identical ($I_1 = I_2 = I_0$): $I = I_0 + I_0 + 2 I_0 \cos \Phi = 2 I_0 (1 + \cos \Phi) = 4 I_0 \cos^2$(Φ divided by 2). Maximum intensity (cos Φ = 1): I max = 4 I_0. Minimum intensity (cos Φ = -1): I min = 0. [1.5 Marks]

• **Key Exam Result:** $I = I_1 + I_2 + 2\text{sqrt}(I_1 I_2) \cos \Phi$; For equal sources: $I = 4 I_0 \cos^2$(Φ divided by 2)

6. What happens to interference fringe width when apparatus is immersed in water? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Effect of Medium on Wavelength]** When Young's double slit apparatus is immersed in water of refractive index n = 4 divided by 3, the frequency ν of light remains invariant, but speed and wavelength decrease: λ water = λ air divided by n = λ air divided by (4 divided by 3) = (three-fourths (0.75)) λ air. [1 Mark]

• **[Step 2: Effect on Fringe Width]** Fringe width is β = λ D divided by d. β water = λ water D divided by d = (λ air divided by n) D divided by d = β air divided by n. β water = β air divided by (4 divided by 3) = (three-fourths (0.75)) β air = 0.75 β air. The fringe width DECREASES by a factor of 1 divided by n (fringes become 25% narrower and closer together). [1 Mark]

• **Key Exam Result:** β water = β air divided by n = (three-fourths (0.75)) β air; Fringe width decreases

7. Numerical on Young's double slit experiment fringe width and angular width [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Data & Formulas]** Slit separation d = 0.28 mm = 2.8×10^{-4} m, screen distance D = 1.4 m, wavelength λ = 600 nm = 6.0×10^{-7} m. [1 Mark]

• **[Step 2: Fringe Width & Angular Width]** 1. Fringe width: β = λ D divided by d = ($6.0 \times 10^{-7} \times 1.4$) divided by ($2.8 \times 10^{-4}$) = ($8.4 \times 10^{-7}$) divided by ($2.8 \times 10^{-4}$) = 3.0×10^{-3} m = 3.0 mm. 2. Distance of 4th bright fringe from center: $y_4 = 4\ \beta = 4 \times 3.0$ mm = 12.0 mm. 3. Angular fringe width: θ = λ divided by d = (6.0×10^{-7}) divided by (2.8×10^{-4}) = 2.14×10^{-3} rad = 0.123°. [2 Marks]

• **Key Exam Result:** Fringe width β = 3.0 mm; y_4 = 12.0 mm; Angular width θ = 2.14 mrad